
Microsoft Excel Training

PIVOT TABLES, CHARTS, SLICERS & DASHBOARDS

Objectives

- Excel Table Theory
- Filters & Sorts
- Pivot Tables & Charts
- Slicers
- Dashboards

EXERCISE

Answer the following questions

1. How many sales representatives are there?
2. How much in sales did representative James Edward make?
3. Which Parish has the highest sales amount?
4. How many days did SERHA have transactions?
5. What is Genuine System Total Sales for 2015?
6. What was the highest selling price for each product?
7. Show a breakdown of sales for each region by parish.
8. Show what each customer paid for each product.

Excel Table Theory



What is an Excel Table

- An Excel Table is a container that holds data
 - making it easier to manage and analyze group of related data, as well as
 - making it easier to use, share and update.

How to Create an Excel Table

1. Select a cell with the data
2. Click on the insert menu and then select Table or hold down the ctrl key on the keyboard and press T
 - In the create Table dialog box, make sure “*my Table has headers*” is checked
3. Click on ok to create the table

Parts of an Excel Table

Column Header Row

- It is the first row in a table and contains the column headings that identify each column of data. Every table column has filtering enabled in the header row so that you can filter or sort your table data quickly.
- Column headings must be unique in the table, they cannot be blank and they cannot contain formulas.
- Column headings remain visible during scrolling large dataset

Excel interface showing a spreadsheet titled "Data Set.xlsx". The ribbon includes File, Home, Insert, Page Layout, Formulas, Data, Review, View, Help, Power Pivot, and Tell me what you want to do. The Home tab is active, displaying options for Clipboard, Font, Alignment, Number, Styles, Cells, and Editing.

The spreadsheet data is as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Transaction ID	Transaction Date	Region	Rep Name	Product	Customer Name	Parish	Sale Amount						
2	1	1/3/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	5100						
3	2	1/14/2014	WRHA	William Able	Basic Analytics Platform	ABC Incorporated	Westmoreland	5100						
4	3	1/15/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	5200						
5	4	1/25/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Hanover	5200						
6	5	3/24/2014	NERHA	James Edwards	Basic Analytics Platform	Advanced Solutions	Portland	5400						
7	6	4/3/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Catherine	5500						
8	7	4/12/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Westmoreland	5500						
9	8	4/12/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	Portland	5600						
10	9	4/20/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Thomas	5600						
11	10	4/23/2014	WRHA	William Able	Basic Analytics Platform	ABC Incorporated	Westmoreland	5700						
12	11	4/30/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	St Catherine	5800						
13	12	5/10/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	St James	5800						
14	13	5/22/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	Kingston & St Andrew	5900						
15	14	5/25/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Thomas	5900						
16	15	5/26/2014	NERHA	James Edwards	Basic Analytics Platform	Advanced Solutions	St Mary	6000						
17	16	5/31/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	6000						
18	17	7/1/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Catherine	6000						
19	18	7/3/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Trelawny	6000						
20	19	7/13/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	Kingston & St Andrew	6100						

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Parts of an Excel Table

Banded Row

- Alternate shading or banding in rows helps to better distinguish the data.

Excel interface showing a spreadsheet titled "Data Set.xlsx". The ribbon includes File, Home, Insert, Page Layout, Formulas, Data, Review, View, Help, Power Pivot, and Tell me what you want to do. The Home ribbon is active, showing options for Clipboard, Font, Alignment, Number, Styles, Cells, and Editing.

The spreadsheet displays a table with the following columns: Transaction ID, Transaction Date, Region, Rep Name, Product, Customer Name, Parish, and Sale Amount. The data is organized into rows, with the first row (row 1) serving as the header.

Transaction ID	Transaction Date	Region	Rep Name	Product	Customer Name	Parish	Sale Amount
1	1/3/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	5100
2	1/14/2014	WRHA	William Able	Basic Analytics Platform	ABC Incorporated	Westmoreland	5100
3	1/15/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	5200
4	1/25/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Hanover	5200
5	3/24/2014	NERHA	James Edwards	Basic Analytics Platform	Advanced Solutions	Portland	5400
6	4/3/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Catherine	5500
7	4/12/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Westmoreland	5500
8	4/12/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	Portland	5600
9	4/20/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Thomas	5600
10	4/23/2014	WRHA	William Able	Basic Analytics Platform	ABC Incorporated	Westmoreland	5700
11	4/30/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	St Catherine	5800
12	5/10/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	St James	5800
13	5/22/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	Kingston & St Andrew	5900
14	5/25/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Thomas	5900
15	5/26/2014	NERHA	James Edwards	Basic Analytics Platform	Advanced Solutions	St Mary	6000
16	5/31/2014	NERHA	James Edwards	Basic Analytics Platform	Genuine Systems	St Ann	6000
17	7/1/2014	SERHA	Susan Smith	Basic Analytics Platform	New York Financial	St Catherine	6000
18	7/3/2014	WRHA	Steven Gross	Basic Analytics Platform	Credit Partners	Trelawny	6000
19	7/13/2014	SERHA	Susan Smith	Basic Analytics Platform	Consumer Technologies Inc	Kingston & St Andrew	6100

The bottom status bar shows "Ready" and "Accessibility: Good to go". The taskbar at the bottom displays various application icons, including Windows, File Explorer, Microsoft Edge, and others. The system clock indicates 6:41 PM on Monday, 4/11/2022.

Parts of an Excel Table

Total Row

- By default, tables don't include a total row but this feature can be enabled if desired. If it's enabled, it will be the last row of the table. This row can contain text, formula or remain blank. Each cell in the total row will have a drop down menu that allows selection of various summary formula.

File

Home

Insert

Page Layout

Formulas

Data

Review

View

Help

Power Pivot

Table Tools

Table Design

Tell me what you want to do

Table Name: Table1

Resize Table

Summarize with PivotTable

Remove Duplicates

Convert to Range

Insert Slicer

Export

Refresh

Unlink

Properties

Open in Browser

Header Row

Total Row

Banded Rows

First Column

Last Column

Banded Columns

Filter Button

Table Style Options

Table Styles

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Portland

	Transaction ID	Transaction Date	Region	Rep Name	Product	Customer Name	Parish	Sale Amount	I	J	K	L	M
148	147	11/9/2014	WRHA	William Able	Cloud Computing Package	ABC Incorporated	St James	21100					
149	148	11/11/2014	NERHA	James Edwards	Cloud Computing Package	Advanced Solutions	St Ann	21200					
150	149	11/13/2014	SERHA	Susan Smith	Cloud Computing Package	New York Financial	St Thomas	21200					
151	150	12/7/2014	WRHA	Steven Gross	Cloud Computing Package	Credit Partners	Hanover	21200					
152	151	12/10/2014	WRHA	Steven Gross	Cloud Computing Package	Credit Partners	Westmoreland	21300					
153	152	12/13/2014	WRHA	William Able	Cloud Computing Package	ABC Incorporated	St James	21400					
154	153	12/22/2014	SERHA	Susan Smith	Cloud Computing Package	New York Financial	St Catherine	21500					
155	154	12/26/2014	NERHA	James Edwards	Cloud Computing Package	Genuine Systems	Portland	21600					
156	155	12/31/2014	SERHA	Susan Smith	Cloud Computing Package	Consumer Technologies Inc	Kingston & St Andrew	21600					
157	156	12/31/2014	SRHA	Mark Edwards	Cloud Computing Package	Western United Bank	Manchester	21600					
158	157	1/8/2015	NERHA	James Edwards	Cloud Computing Package	Advanced Solutions	Portland	21700					
159	158	1/21/2015	SERHA	Susan Smith	Cloud Computing Package	New York Financial	Kingston & St Andrew	21700					
160	159	1/26/2015	WRHA	William Able	Cloud Computing Package	ABC Incorporated	Trelawny	21700					
161	160	2/16/2015	WRHA	William Able	Cloud Computing Package	ABC Incorporated	St James	21900					
162	161	2/20/2015	NERHA	James Edwards	Cloud Computing Package	Advanced Solutions	Portland	22000					
163	162	2/21/2015	SERHA	Susan Smith	Cloud Computing Package	Consumer Technologies Inc	St Catherine	22000					
164	163	3/8/2015	NERHA	James Edwards	Cloud Computing Package	Advanced Solutions	St Mary	22500					
165	Total							2363000					
166													
167													

Ready

Accessibility: Investigate

6:45 PM

Monday

4/11/2022

Why use Excel Data Table

- Tables are very easily formatted
- Table headers remain visible even when you scroll down
- Filters are added to your data
- Tables automatically expand when you add new rows, columns or formulas including their formats
- You can add Totals or other mathematical functions to Tables automatically
- You can create dynamic charts and add slicers and timelines to filter the data and charts

Filtering & Sorting



Filtering & Sorting

- Sorting is putting the data in a particular order.
- Filter is viewing only a portion of the data that you are interested in based on one or more criteria.

EXERCISE



DATA TABLE

EXERCISE

Create a Data Table from the provided dataset and answer the following questions

1. How many sales representatives are there?
2. How much in sales did representative James Edward make?
3. Which Parish has the highest sales amount?
4. What was the highest selling price for each product?
5. Show a breakdown of sales for each region by parish.
6. Show what each customer paid for each product.

EXERCISE

Create a Table from the provided dataset.

1. How many facilities are there?
2. Which facility had the largest physical inventory?
3. Which parish is Darlinston Health Center located?

Pivot Table & Chart



Pivot Table

- A pivot table is a program tool that allows you to reorganize and summarize selected columns and rows of data in Microsoft excel to obtain a desired report. A pivot table doesn't actually change the spreadsheet itself. In database lingo, to pivot is to turn the data to view it from different perspectives.

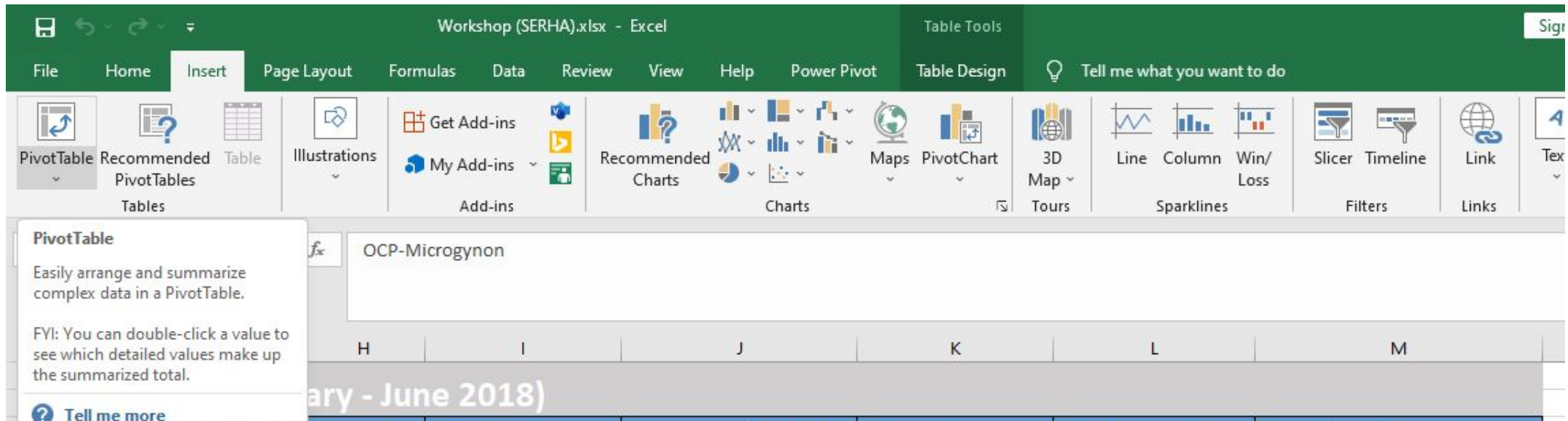


Pivot Table allows you to:

- Count the data
- Filter data on one or more fields
- Calculate totals, averages, running totals, percentage distributions, minimums, and more, all with a few simple clicks and without writing any formula.
- Summarize data by one or more variables
- Organize daily data into months and years, and easily graph the result
- Count things in the data e.g. How many employee in a department

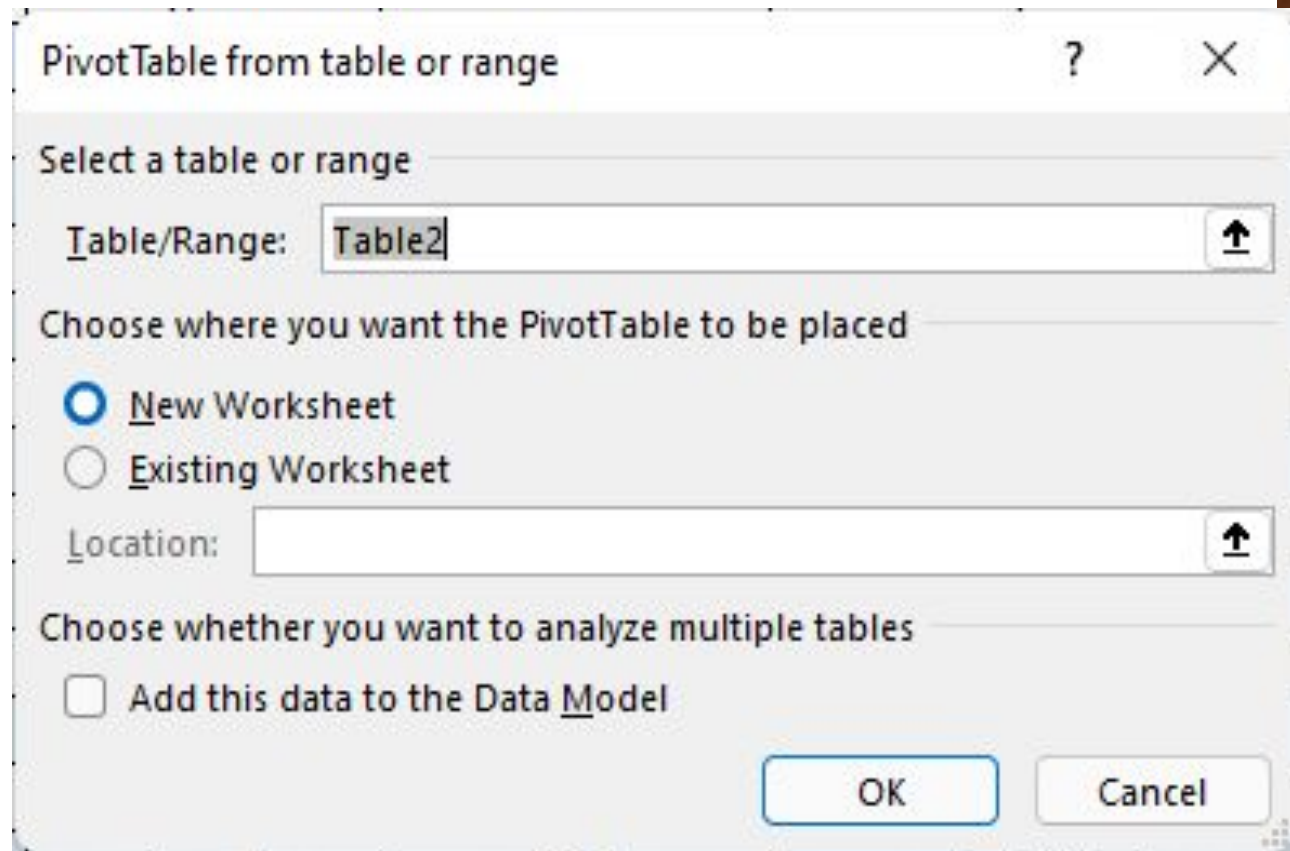
How to insert a Pivot Table:

- Make sure to select a single cell in your data/table
- Click on the insert menu and then select PivotTable



How to insert a Pivot Table:

- In the PivotTable dialog box shown make sure
 - The correct Table or Range is selected
 - New Worksheet is selected
 - If you want to count unique values then
 - Select “Add this data to the Data Model”
 - Click okay



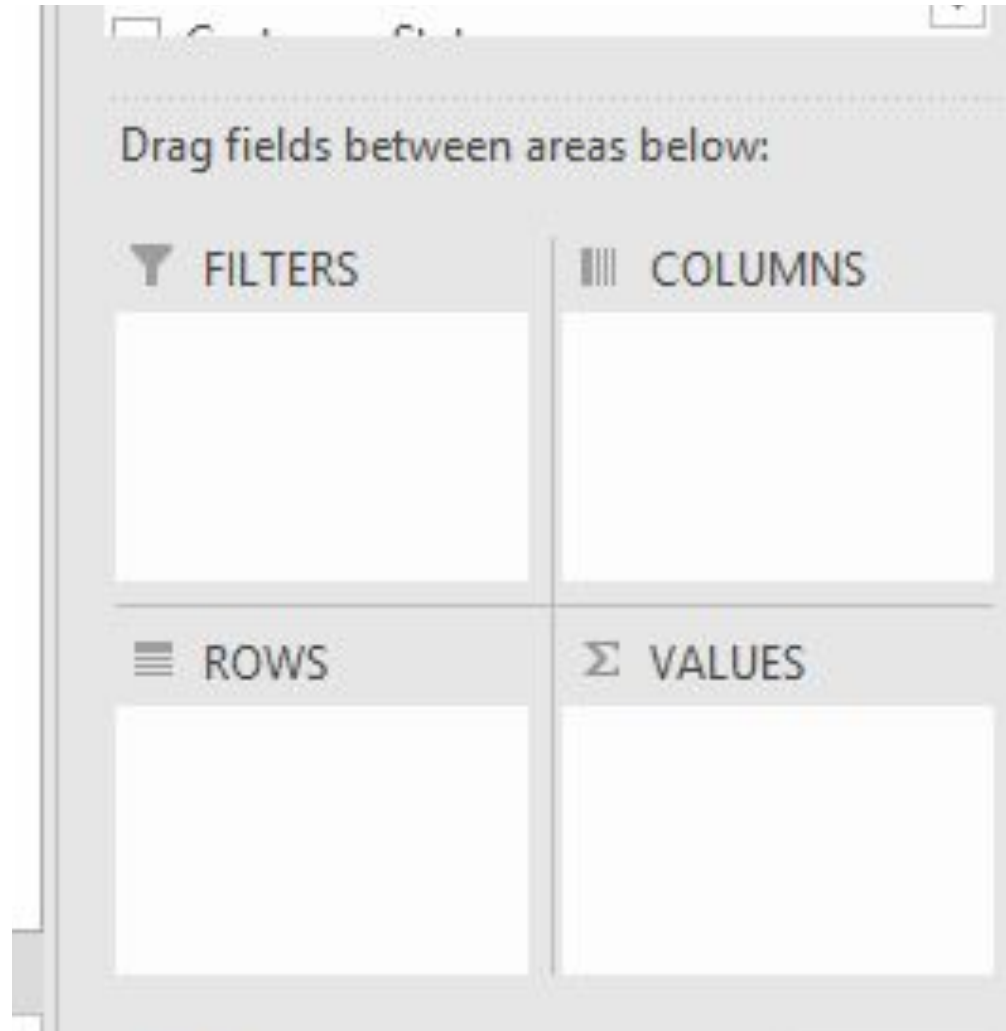
The screenshot shows the 'PivotTable from table or range' dialog box. The title bar reads 'PivotTable from table or range'. The first section, 'Select a table or range', has a text box labeled 'Table/Range:' containing the text 'Table2'. The second section, 'Choose where you want the PivotTable to be placed', has two radio buttons: 'New Worksheet' (which is selected) and 'Existing Worksheet'. Below these is a text box labeled 'Location:'. The third section, 'Choose whether you want to analyze multiple tables', has a checkbox labeled 'Add this data to the Data Model' which is currently unchecked. At the bottom right are 'OK' and 'Cancel' buttons.

Show

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Part of Pivot Table:



Part of Pivot Table:

- **Rows** - Drag a field to the rows section, and it shows up vertically in the pivot table.
- **Column** - Columns work the same way as rows. Summarize data by columns along the top instead of along the left.
- **Filter** - These allow you to restrict the entire data set based on the value of one or more fields.
- **Value** – This allows you to see the result using one of several functions. Either count, sum, average, max, min etc.

You can have multiple fields within a pivot table area. You can also create two-dimensional pivot tables by using one field along the rows and another field along the columns.

Ways to filter a Pivot Table:

- **Filter** – Drag a field you want to filter into the filter box and then choose what should be filtered.
- **Filter on the row/column** – click on the column/row header you want filtered and select filter
- **Use a Slicer** – Insert a Slicer and use it to filter the data

EXERCISE



PIVOT TABLE

EXERCISE

Create PivotTables from the provided dataset and answer the following questions

1. How many sales representatives are there?
2. How much in sales did representative James Edward make?
3. Which Parish has the highest sales amount?
4. What was the highest selling price for each product?
5. Show a breakdown of sales for each region by parish.
6. Show what each customer paid for each product.

EXERCISE

CREATE PIVOT TABLES FOR THE FOLLOWING QUESTIONS.

- 1) Use a pivot table to show the amount of expired quantity for each Facility?
Rename the worksheet “*Expired Quantity*”
- 2) Using a Pivot Table, create a table displaying the amount of contraceptive that were used for the period for each facility. Rename the worksheet “*Contraceptive Amount*”
- 3) Create a Pivot Table displaying the physical inventory for each facility. Rename the worksheet “*Physical Inventory*”
- 4) Create a Pivot Table to show the facilities with days stocked out. Rename the worksheet “*Stockout Days*”
- 5) Create a Pivot Table to show the facilities with their numbers of stocked out. Rename the worksheet “*Number of Stockout*”

Pivot Chart

- Pivot Charts are a wonderful way to visualize data in an effective and efficient manner and is a visual representation of a Pivot Table. A pivot chart is especially useful for user when dealing with tremendous amounts of data.

How to insert a Pivot Chart

Create a PivotChart

- **Select a cell in your table.**
- **Select Insert > PivotChart PivotChart option on the ribbon.**
- **Select OK.**

Create a chart from a PivotTable

- **Select a cell in your PivotTable.**
- **Select PivotTable Tools > Analyze > PivotChart PivotChart option on the ribbon.**
- **Select a chart.**
- **Select OK.**

EXERCISE

Create Pivot Charts from the PivotTables created earlier

1. How many sales representatives are there?
2. How much in sales did representative James Edward make?
3. Which Parish has the highest sales amount?
4. What was the highest selling price for each product?
5. Show a breakdown of sales for each region by parish.
6. Show what each customer paid for each product.

EXERCISE

CREATE PIVOT CHARTS FOR THE FOLLOWING WORKSHEETS.

- 1) *“Expired Quantity”*
- 2) *“Contraceptive Amount”*
- 3) *“Physical Inventory”*
- 4) *“Stockout Days”*
- 5) *“Number of Stockout”*

Slicers

- Slicers are visual filters. Using a slicer, you can filter your data (or pivot table, pivot chart) by clicking on the type of data you want.
- In addition to quick filtering, slicers also indicate the current filtering state, which makes it easy to understand what exactly is currently displayed.

How to insert a Slicer

- Click anywhere in the table or PivotTable.
- On the Home tab, go to Insert > Slicer.
- Insert Slicer
- In the Insert Slicers dialog box, select the check boxes for the fields you want to display, then select OK.
- A slicer will be created for every field that you selected. Clicking any of the slicer buttons will automatically apply that filter to the linked table or PivotTable.
- If you want to connect a slicer to more than one PivotTable, go to Slicer > Report Connections > check the PivotTables to include, then select OK.

Parts of a Slicer



1. A slicer header indicates the category of the items in the slicer.
2. A filtering button that is not selected indicates that the item is not included in the filter.
3. A filtering button that is selected indicates that the item is included in the filter.
4. A Clear Filter button removes the filter by selecting all items in the slicer.
5. A scroll bar enables scrolling when there are more items than are currently visible in the slicer.
6. Border moving and resizing controls allow you to change the size and location of the slicer.

EXERCISE

INSERT SLICERS FOR THE FOLLOWING.

- 1) Sales Representative
- 2) Region
- 3) Quarters

EXERCISE

INSERT SLICERS FOR THE FOLLOWING.

- 1) Parish**
- 2) Facility Type**
- 3) Health District**
- 4) Family Planning Method**

Dashboard

- Dashboards track KPIs, metrics, and other data points in one visual, central place. They give you a high-level view of work, helping you make quick decisions and keeping everyone up to date. A dashboard's visual nature simplifies complex data and provides an at-a-glance view of current status or performance in real time.

EXERCISE



DASHBOARD

EXERCISE

CREATE A NEW WORKSHEET NAMED DASHBOARD.

- 1) Add all Pivot Charts to this worksheet**
- 2) Add all Slicers to this worksheet**
 - 1) Ensure that all slicers are connected to all Pivot Tables**
- 3) What was the highest selling price for each product?**
- 4. Show a breakdown of sales for each region by parish.**
- 5. Show what each customer paid for each product.**

EXERCISE

INSERT THE PIVOT CHARTS & SLICERS IN A NEW WORKSHEET.

ANSWER THE FOLLOWING

- 1) Which facility had the most total stock out?**
- 2) Which Parish had the lowest expired quantity?**
- 3) Which Facility Type had the lowest physical inventory?**
- 4) Which Health District had the most stock out?**
- 5) Which facility saw the most stock out days?**